

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Digital Campus Library Reservation Gateway

1. Architecture Selection

We chose Microservices Architecture for the Digital Campus Library Reservation Gateway.

The system is divided into independent services for catalog search, reservation/hold queue management, notifications, and fine management, accessed through an API Gateway.

2. Architectural Choice – Reasons

- Independent scalability: The Catalog Search Service can be independently optimized and scaled to meet the requirement of returning results for 50,000 titles in less than 150 ms, without scaling unrelated services.
- Separation of responsibilities: Search, FIFO hold queues, holding notifications, and overdue fine transactions are separated into services, making the system easier to maintain and allowing services to be developed or deployed independently.

3. Security Advantage

The API Gateway provides a centralized point for authentication, authorization, input validation, and request control before requests reach internal services. This helps protect library and user data and restrict unauthorized access to reservation and fine operations.

4. Performance Benefit

Microservices allow performance-critical operations to be isolated. In particular, the Catalog Search Service can be optimized and scaled independently for fast ISBN/catalog searches under peak load, helping the system target the required response time of less than 150 ms for 50,000 titles.

5. Main Components and Interactions

The component model contains the Library User Interface, API Gateway, Catalog Search Service, Reservation/Hold Queue Service, Notification Service, Fine Management Service, and Library Database. The main interfaces include User Interface API, Catalog Search API, Reservation API, Notification Interface, Fine Management API, and database interfaces. The Reservation/Hold Queue Service maintains FIFO queue positions and triggers notifications when a book becomes available.

Component Diagram :

